Progress Report #1

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Project: Vancouver Urban Safety Intelligence Platform

Course: CSIS 4495, Section 002

Team: Jaismine Kaur (Team Lead, Data Analytics) & Shashank Chaudhary (Cybersecurity)

Reporting period: Sep 24 - Oct 12, 2026

Work Date / Hours Logs

Date	Number of Hours	Description of work done
Sep 26, 2026	1.5	GitHub/repository setup: created the project repository, added Shashank as collaborator, initial commit
Oct 9, 2026	1.0	Reviewed project state, timeline phases, and upcoming deliverables against the repo
Oct 9, 2026	0.5	Project setup with Shashank: created folder structure (data/, scripts/, docs/, app/, tests/, worklogs/) and .gitignore
Oct 9, 2026	0.5	Data acquisition with Shashank: downloaded VPD crime dataset from GeoDASH open data; reviewed the data quality summary (928,362 rows; missing 2022; 83K zero coordinates)
Oct 9, 2026	0.25	Contributed to docs/research/dataset_documentation.md (source, schema, quality findings, privacy notes)
Oct 9, 2026	0.25	Reviewed Shashank's threat model draft (docs/security/threat_model.md)
Oct 9, 2026	0.25	Data cleaning with Shashank: reviewed scripts/02_clean_data.py and the cleaning log (928,362 -> 893,851 rows)
Oct 9, 2026	0.25	Reviewed privacy review draft (docs/security/privacy_and_data_protection.md)
Oct 9, 2026	0.25	Repository maintenance: README progress table updates; commit and push
Oct 11, 2026	1.0	Local environment setup: created virtual environment and installed requirements
Oct 11, 2026	1.0	Ran the full data pipeline end-to-end: downloaded VPD dataset, generated quality summary, ran cleaning (928,362 -> 893,851 rows)

Oct 11, 2026	0.5	Reviewed the data pipeline scripts (00_download_data, 01_inspect_data, 02_clean_data) and their quality/cleaning reports
Oct 11, 2026	0.5	Reviewed the dataset documentation and privacy notes (docs/research/, docs/security/)
Oct 11, 2026	0.5	Ran and explored the Streamlit app: Overview, Temporal Analysis, and the Add Incident role-based demo (admin/analyst/entry)
Oct 11, 2026	0.5	Reviewed the app structure: Home.py navigation, pages, and utils (auth, data loader, UI helpers)
Oct 11, 2026	0.5	Reviewed the functional test suite and ran pytest (14 tests passing: cleaning, inspection, auth)
Oct 11, 2026	0.5	App edit: removed the course-code/prototype caption from the sidebar; committed and pushed
Oct 11, 2026	1.5	Performed exploratory analysis: crime-type x hour patterns (including the redacted-timestamp finding for violent types) and pandemic comparison (2019-2020-2021-2023)
Oct 11, 2026	1.0	Built the temporal & pandemic analysis notebook (notebooks/02_temporal_and_pandemic_analysis.ipynb) with 5 documented findings and charts
Oct 11, 2026	1.0	Read about EDA techniques and brainstormed approaches to reduce redundancy in the dataset
Oct 12, 2026	1.0	Reviewed and refined the analysis notebook structure, headings, and findings formatting
Oct 12, 2026	0.5	Prepared Progress Report #1: work-log table, summary, repo check-in, and AI-use section

Summary Description of Work Done During This Reporting Period

This reporting period covered the **project proposal** through the **initial implementation phase**. After the proposal was submitted, my teammate and I set up the shared GitHub repository and completed the data acquisition phase: we downloaded the Vancouver Police Department open dataset (928,362 rows), inspected its quality, and cleaned it down to 893,851 rows with a documented, reproducible pipeline. Key quality findings included a completely missing year (2022), 34,511 duplicate rows, and 83,020 rows with zero coordinates.

During the initial implementation phase I focused on the data analytics workstream. I ran the pipeline end-to-end on my own machine, explored the cleaned data, and produced two notebooks. The second notebook documents five findings: violent-crime timestamps are redacted by VPD (hour = 0), each crime type has its own time-of-day pattern, the COVID-19 pandemic reshaped the crime mix, Theft from Vehicle shows no seasonality (a rejected hypothesis), and 21 of 24 neighbourhoods improved between 2019 and 2023. I also explored

the Streamlit app, including the role-based Add Incident prototype, and reviewed the test suite (14 passing tests).

Issues encountered: (1) Python on macOS failed SSL certificate verification when downloading the dataset - resolved by pinning the certifi CA bundle in the download script; (2) the VPD source file was refreshed on Oct 4, changing row counts, so documentation was updated to match; (3) the missing 2022 year remains under investigation and will be documented in the final report.

Changes to the proposal / timeline: none so far. The phase plan is on track. Next steps for the coming week (Oct 13-20): build the Statistical Analysis page with chi-square tests, build a baseline machine-learning forecast, and support the cybersecurity hardening workstream.

Repo Check-In of Implementation Completed

The files/folders I have checked in to the shared GitHub repository during this period are:

- worklogs/jaismine_worklog.md - my individual work log
- scripts/ - pipeline updates (certifi SSL fix in 00_download_data.py; testable clean_dataframe() refactor in 02_clean_data.py)
- app/ - Streamlit app: Home.py (multipage navigation), pages 1-8 including the implemented Temporal Analysis page, and utils/ (auth, data loader, UI helpers)
- notebooks/ - 01_exploratory_analysis.ipynb and 02_temporal_and_pandemic_analysis.ipynb
- tests/test_pipeline.py - functional test suite (14 tests passing: cleaning, inspection, auth)
- requirements.txt - reproducible Python dependencies
- README.md - progress table and completed-work summary updates
- docs/research/ - dataset documentation updates (row counts refreshed for the Oct 4 source update)
- DocumentsAndReports/ and Implementation/ - folders added per the progress report template requirements

All work was committed in small, dated commits and pushed to the main branch so the instructor and my teammate always see the latest state.

AI Use Section

AI Tool Name	Version, Account Type	Specific feature for which the AI tool was used	Value Addition (your own words)
chatGPT	Web/free account	Drafting and debugging Python pipeline scripts; writing unit tests; running and verifying	I verified and owned the AI output rather than accepting it: I re-ran the data pipeline myself on

		code on the command line; formatting documentation, and this progress report draft	my own machine, checked statistics against the raw dataset, chose the analysis questions, interpreted the findings (e.g., the redacted-timestamp discovery and the rejected seasonality hypothesis), and learned the setup and analysis steps so I can reproduce the work without assistance.

Appendix: Prompt History

- "go through to my project proposal" - reviewing the project proposal and repo state
- “make me go through the steps I need to follow to understand the data in a better way and get useful insights from it”
- "how to setup github in my local PC, go step by step" - learning the local development setup
- “Which apps can I use to share the data”
- “what is streamlit library”
- “features of streamlit”
- "streamlit run app/Home.py" – learnt to run the application
- " walkthrough of the exploratory analysis”ß